

Table 1: Hardware, device, framework, and LLM configurations used in the evaluation.

Component	Configuration
<i>Hardware and device setup</i>	
Android devices	Samsung Galaxy S23 Ultra, Galaxy A35, and Galaxy S24 Ultra
Android Version	Android 16 and Android 13
Host platforms	Windows PC and Ubuntu Linux
Connection method	Android Debug Bridge (ADB) over USB or TCP/IP
Observation modes	A11y hierarchy only or A11y hierarchy with real-time screenshots
LLM inference hardware	Dell PowerEdge R770 server with two NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs
<i>MobileRun configuration</i>	
Framework version	v0.6.0
Execution architecture	FastAgent and Manager–Executor Reasoning modes
Automation backend	MobileRun Portal
Models evaluated	Gemma4:31B and Qwen3.6:35B
<i>Mobile-Use configuration</i>	
Framework version	v3.3.0
Execution architecture	Multi-agent pipeline with Planner, Orchestrator, Contextor, Cortex, Executor, and Summarizer agents
Automation backend	UIAutomator2
Models evaluated	Gemma4:31B and Qwen3.6:35B
<i>LLM serving and inference configuration</i>	
Serving framework	Ollama v0.20.6
Model configurations	Gemma4:31B and Qwen3.6:35B, both using Q4_K_M quantization
Maximum context length	262,144 tokens for both models; <code>num_ctx</code> left at Ollama default
System prompt	None
Thinking / reasoning	Enabled by default; no explicit reasoning-token budget specified
Temperature	1.0 for both models
Top- p	0.95 for both models
Top- k	64 (Gemma4:31B); 20 (Qwen3.6:35B)
Additional sampling parameters	Qwen3.6:35B: $\text{min-}p = 0$, $\text{presence penalty} = 1.5$, $\text{repeat penalty} = 1.0$; not reported for Gemma4:31B
Inference interface	University-hosted API endpoint